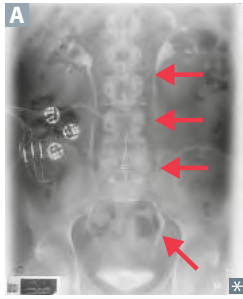


Course of ureters

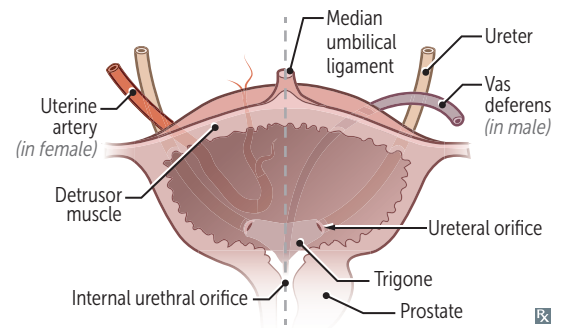
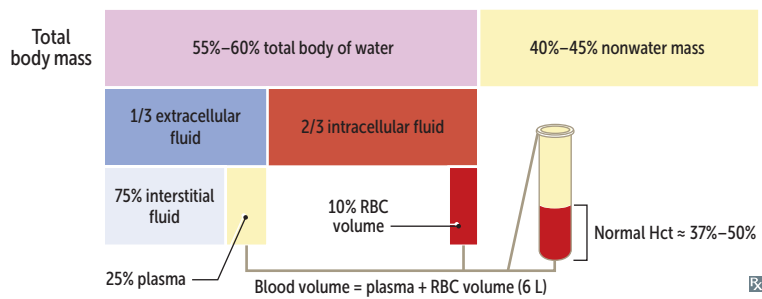
Course of ureter **A**: arises from renal pelvis, travels under gonadal arteries → **over** common iliac artery → **under** uterine artery/vas deferens (retroperitoneal).

Gynecologic procedures (eg, ligation of uterine or ovarian vessels) may damage ureter → ureteral obstruction or leak.

Bladder contraction compresses the intramural ureter, preventing urine reflux.

3 common points of ureteral obstruction: ureteropelvic junction, pelvic inlet, ureterovesical junction.

Water (ureters) flows **over** the iliacs and **under** the bridge (uterine artery or vas deferens).

**► RENAL—PHYSIOLOGY****Fluid compartments**

HIKIN: **HI** K^+ **IN**tracellularly.

60–40–20 rule (% of body weight for average person):

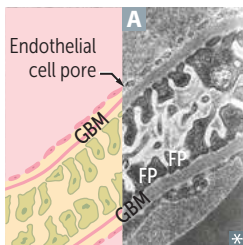
- 60% total body water
- 40% ICF, mainly composed of K^+ , Mg^{2+} , organic phosphates (eg, ATP)
- 20% ECF, mainly composed of Na^+ , Cl^- , HCO_3^- , albumin

Plasma volume can be measured by radiolabeling albumin.

Extracellular volume can be measured by inulin or mannitol.

Serum osmolality = 275–295 mOsm/kg H_2O .

Plasma volume = $TBV \times (1 - Hct)$.

Glomerular filtration barrier

Responsible for filtration of plasma according to size and charge selectivity.

Composed of

- Fenestrated capillary endothelium
- Glomerular basement membrane (GBM) with type IV collagen chains and heparan sulfate
- Visceral epithelial layer consisting of podocyte foot processes (FPs) **A**

Charge barrier—glomerular filtration barrier contains $\ominus$ charged glycoproteins that prevent entry of $\ominus$ charged molecules (eg, albumin).

Size barrier—fenestrated capillary endothelium (prevents entry of > 100 nm molecules/blood cells); slit diaphragm (prevents entry of molecules > 40 – 50 nm).

Podocyte foot processes interpose with GBM—both charge and size barrier.

Renal clearance

$C_x = (U_x V)/P_x$ = volume of plasma from which the substance is completely cleared in the urine per unit time.

If $C_x < \text{GFR}$: net tubular reabsorption and/or not freely filtered.

If $C_x > \text{GFR}$: net tubular secretion of X.

If $C_x = \text{GFR}$: no net secretion or reabsorption.

C_x = clearance of X (mL/min).

U_x = urine concentration of X (eg, mg/mL).

P_x = plasma concentration of X (eg, mg/mL).

V = urine flow rate (mL/min).

Glomerular filtration rate

Inulin clearance can be used to calculate GFR because it is freely filtered and is neither reabsorbed nor secreted.

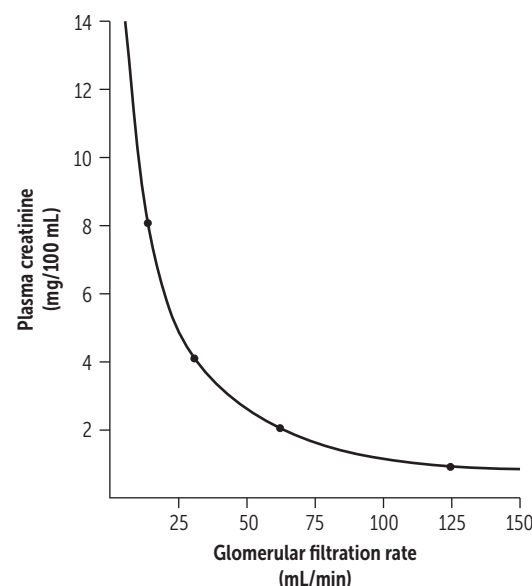
$$C_{\text{inulin}} = \text{GFR} = U_{\text{inulin}} \times V / P_{\text{inulin}} \\ = K_f [(P_{\text{GC}} - P_{\text{BS}}) - (\pi_{\text{GC}} - \pi_{\text{BS}})]$$

P_{GC} = glomerular capillary hydrostatic pressure;

P_{BS} = Bowman space hydrostatic pressure; π_{GC} = glomerular capillary oncotic pressure; π_{BS} = Bowman space oncotic pressure; π_{BS} normally equals zero; K_f = filtration coefficient.

Normal GFR $\approx$ 100 mL/min.

Creatinine clearance is an approximate measure of GFR. Slightly overestimates GFR because creatinine is moderately secreted by proximal renal tubules.

**Renal blood flow autoregulation**

Autoregulatory mechanisms help maintain a constant RBF and GFR to protect the kidney from rapid fluctuations in renal perfusion pressure that could cause renal injury leading to reduced glomerular filtration. Mechanisms:

Myogenic: $\uparrow$ arterial pressure $\rightarrow$ stretch of afferent arteriole $\rightarrow$ mechanical activation of vascular smooth muscle $\rightarrow$ vasoconstriction of afferent arteriole $\rightarrow \downarrow$ RBF.

Tubuloglomerular: $\uparrow$ NaCl of the filtrate sensed by macula densa cells $\rightarrow$ paracrine-driven vasoconstriction of afferent arteriole $\rightarrow \downarrow$ RBF. Can also have the opposite effect on RBF if $\downarrow$ NaCl.

Effective renal plasma flow

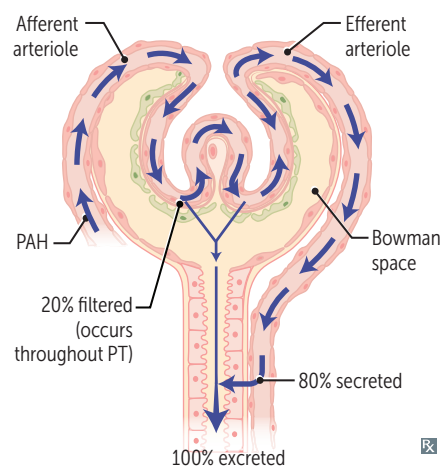
Effective renal plasma flow (eRPF) can be estimated using *para*-aminohippuric acid (PAH) clearance. Between filtration and secretion, there is nearly complete excretion of all PAH that enters the kidney.

$$e\text{RPF} = U_{\text{PAH}} \times V / P_{\text{PAH}} = C_{\text{PAH}}$$

$$\text{Renal blood flow (RBF)} = \text{RPF} / (1 - \text{Hct}).$$

Usually 20–25% of cardiac output.

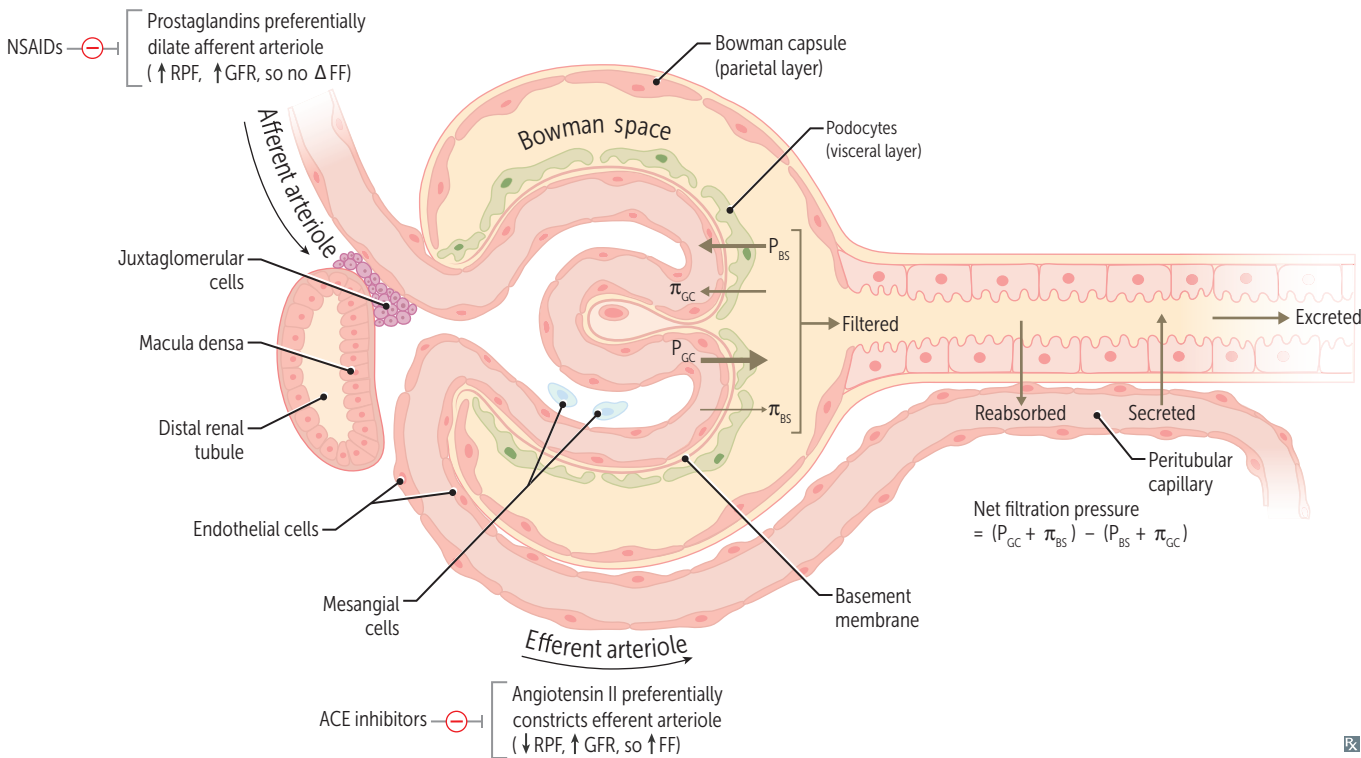
eRPF underestimates true renal plasma flow (RPF) slightly.



Filtration

Filtration fraction (FF) = GFR/RPF .
Normal FF = 20%.
Filtered load (mg/min) = $GFR \text{ (mL/min)} \times \text{plasma concentration (mg/mL)}$.

GFR can be estimated with creatinine clearance.
Prostaglandins Dilate Afferent arteriole (PDA).
Angiotensin II Constricts Efferent arteriole (ACE). RPF is best estimated with PAH clearance. NSAIDs and ACE inhibitors should not be given together → constriction of afferent and efferent arterioles.



Changes in glomerular dynamics

	GFR	RPF	FF (GFR/RPF)
Afferent arteriole constriction	$\downarrow$	$\downarrow$	—
Efferent arteriole constriction	$\uparrow$	$\downarrow$	$\uparrow$
$\uparrow$ plasma protein concentration	$\downarrow$	—	$\downarrow$
$\downarrow$ plasma protein concentration	$\uparrow$	—	$\uparrow$
Constriction of ureter	$\downarrow$	—	$\downarrow$
Dehydration	$\downarrow$	$\downarrow\downarrow$	$\uparrow$

Notably for patients undergoing nephrectomy, there is a proportionate decline in renal function ($\downarrow$ nephron number) → $\downarrow$ remaining kidney renal function to ~50% of pre-nephrectomy value until long-term compensations like hypertrophy develop.

Calculation of reabsorption and secretion rate

Filtered load = $GFR \times P_x$.

Excretion rate = $V \times U_x$.

Reabsorption rate = filtered – excreted.

Secretion rate = excreted – filtered.

FENa = fractional excretion of sodium.

$$FENa = \frac{Na^+ \text{ excreted}}{Na^+ \text{ filtered}} = \frac{V \times U_{Na}}{GFR \times P_{Na}} = \frac{P_{Cr} \times U_{Na}}{U_{Cr} \times P_{Na}} \text{ where } GFR = \frac{U_{Cr} \times V}{P_{Cr}}$$

Glucose clearance

Glucose at a normal plasma level (range 60–120 mg/dL) is completely reabsorbed in proximal convoluted tubule (PCT) by Na^+ /glucose cotransport.

In adults, at plasma glucose of ~ 200 mg/dL, glucosuria begins (threshold). At rate of ~ 375 mg/min, all transporters are fully saturated (T_m).

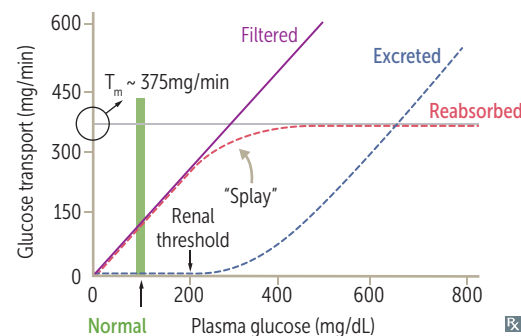
Normal pregnancy is associated with ↑ GFR.

With ↑ filtration of all substances, including glucose, the glucose threshold occurs at lower plasma glucose concentrations → glucosuria at normal plasma glucose levels.

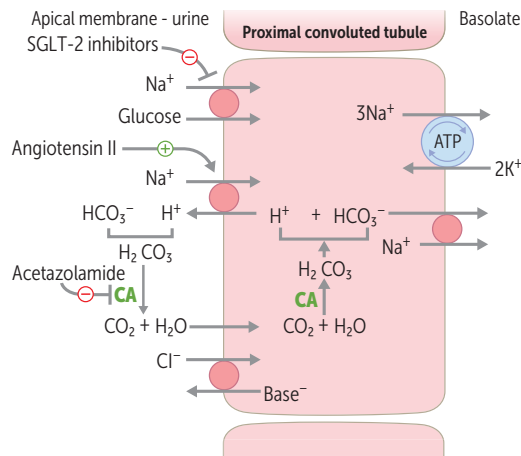
Sodium-glucose cotransporter 2 (SGLT2) inhibitors (eg, -flozin drugs) lead to glucosuria at plasma concentrations < 200 mg/dL.

Glucosuria is an important clinical clue to diabetes mellitus.

Splay phenomenon— T_m for glucose is reached gradually rather than sharply due to the heterogeneity of nephrons (ie, different T_m points); represented by the portion of the titration curve between threshold and T_m .



Nephron transport physiology



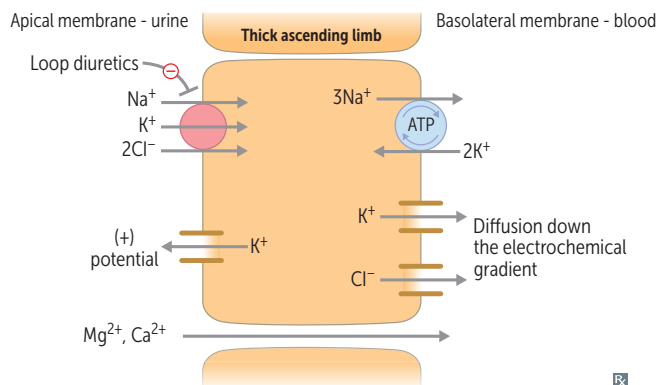
Early PCT—contains brush border. Reabsorbs all glucose and amino acids and most HCO_3^- , Na^+ , Cl^- , PO_4^{3-} , K^+ , H_2O , and uric acid. Isotonic absorption. Generates and secretes NH_3 , which enables the kidney to excrete (via secretion) more H^+ .

PTH—inhibits $\text{Na}^+/\text{PO}_4^{3-}$ cotransport → ↑ PO_4^{3-} excretion.

AT II—stimulates Na^+/H^+ exchange → ↑ Na^+ , H_2O , and HCO_3^- reabsorption (permitting contraction alkalosis). 65–80% Na^+ and H_2O reabsorbed.

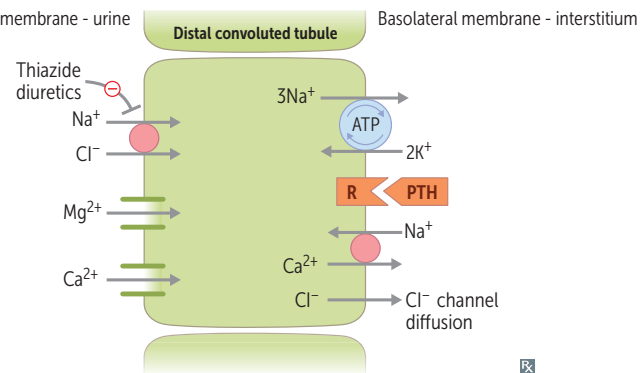
Thin descending loop of Henle—passively reabsorbs H_2O via medullary hypertonicity (impermeable to Na^+).

Concentrating segment. Makes urine hypertonic.



Thick ascending loop of Henle—reabsorbs Na^+ , K^+ , and Cl^- . Indirectly induces paracellular reabsorption of Mg^{2+} and Ca^{2+} through ⊕ lumen potential generated by K^+ backleak. Impermeable to H_2O . Makes urine less concentrated as it ascends.

10–20% Na^+ reabsorbed.

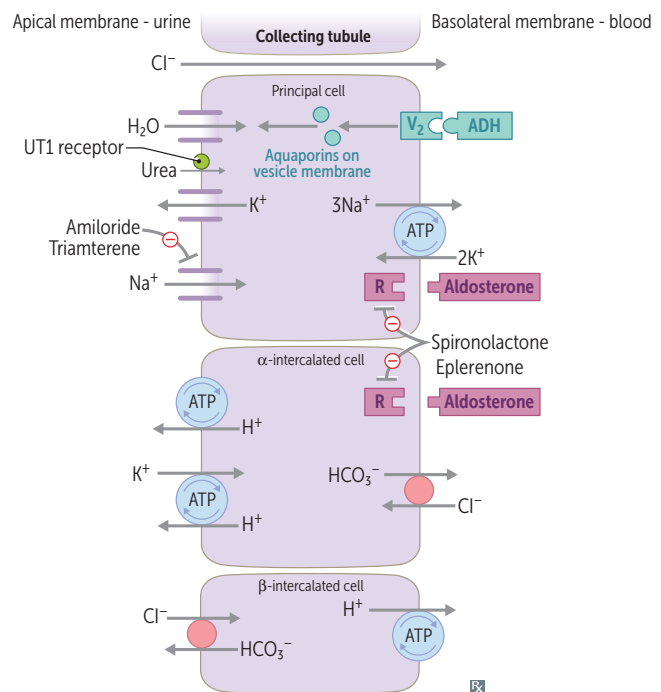


Early DCT—reabsorbs Na^+ , Cl^- . Impermeable to H_2O .

Makes urine fully dilute (hypotonic).

PTH—↑ $\text{Ca}^{2+}/\text{Na}^+$ exchange → ↑ Ca^{2+} reabsorption.

5–10% Na^+ reabsorbed.

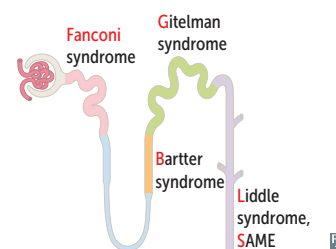


Collecting tubule—reabsorbs Na^+ in exchange for secreting K^+ and H^+ (regulated by aldosterone).

Aldosterone—acts on mineralocorticoid receptor → mRNA → protein synthesis. In principal cells: ↑ apical K^+ conductance, ↑ Na^+/K^+ pump, ↑ epithelial Na^+ channel (ENaC) activity → lumen negativity → K^+ secretion. In α-intercalated cells: lumen negativity → ↑ H^+ ATPase activity → ↑ H^+ secretion → ↑ $\text{HCO}_3^-/\text{Cl}^-$ exchanger activity.

ADH—acts at V_2 receptor → insertion of aquaporin H_2O channels on apical side.

3–5% Na^+ reabsorbed. Urea reabsorption (medullary, not cortical, duct only).

Renal tubular defects Order: **Fanconi's BaGeLS**

	DEFECTS	EFFECTS	CAUSES	NOTES
Fanconi syndrome	Generalized reabsorption defect in PCT → ↑ excretion of amino acids, glucose, HCO_3^- , and PO_4^{3-} , and all substances reabsorbed by the PCT	Metabolic acidosis (proximal RTA), hypophosphatemia, hypokalemia	Hereditary defects (eg, Wilson disease, tyrosinemia, glycogen storage disease), ischemia, multiple myeloma, drugs (eg, ifosfamide, cisplatin, tenofovir, lead poisoning)	Growth restriction and rickets/osteopenia common due to hypophosphatemia Volume depletion also common
Bartter syndrome	Reabsorption defect in thick ascending loop of Henle (affects $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter)	Metabolic alkalosis, hypochloremia, hypokalemia, hypercalciuria	Autosomal recessive	Presents similarly to chronic loop diuretic use
Gitelman syndrome	Reabsorption defect of NaCl in DCT	Metabolic alkalosis, hypochloremia, hypomagnesemia, hypokalemia	Autosomal recessive	Presents similarly to chronic thiazide diuretic use Less severe than Bartter syndrome
Liddle syndrome	Gain of function mutation → ↓ Na^+ channel degradation → ↑ Na^+ reabsorption in collecting tubules	Metabolic alkalosis, hypokalemia, hypertension, ↓ serum aldosterone	Autosomal dominant	Presents similarly to hyperaldosteronism, but aldosterone is nearly undetectable Treatment: amiloride
Syndrome of Apparent Mineralocorticoid Excess	Cortisol activates mineralocorticoid receptors; 11β -HSD converts cortisol to cortisone (inactive on these receptors) Hereditary 11β -HSD deficiency → ↑ cortisol → ↑ mineralocorticoid receptor activity	Metabolic alkalosis, hypokalemia, hypertension ↓ serum aldosterone level; cortisol tries to be the SAME as aldosterone	Autosomal recessive Can acquire disorder from glycyrrhetic acid (present in licorice), which blocks activity of 11β -hydroxysteroid dehydrogenase	Treatment: K^+ -sparing diuretics (↓ mineralocorticoid effects) or corticosteroids (exogenous corticosteroid ↓ endogenous cortisol production → ↓ mineralocorticoid receptor activation)

Features of renal disorders

CONDITION	BLOOD PRESSURE	PLASMA RENIN	ALDOSTERONE	SERUM Mg^{2+}	URINE Ca^{2+}
SIADH	—/↑	—/↓	—/↓	—	—
Bartter syndrome	—	↑	↑	—	↑
Gitelman syndrome	—	↑	↑	↓	↓
Renin-secreting tumor	↑	↑	↑	—	—
Primary hyperaldosteronism	↑	↓	↑	—	—
Liddle syndrome, syndrome of apparent mineralocorticoid excess	↑	↓	↓	—	—

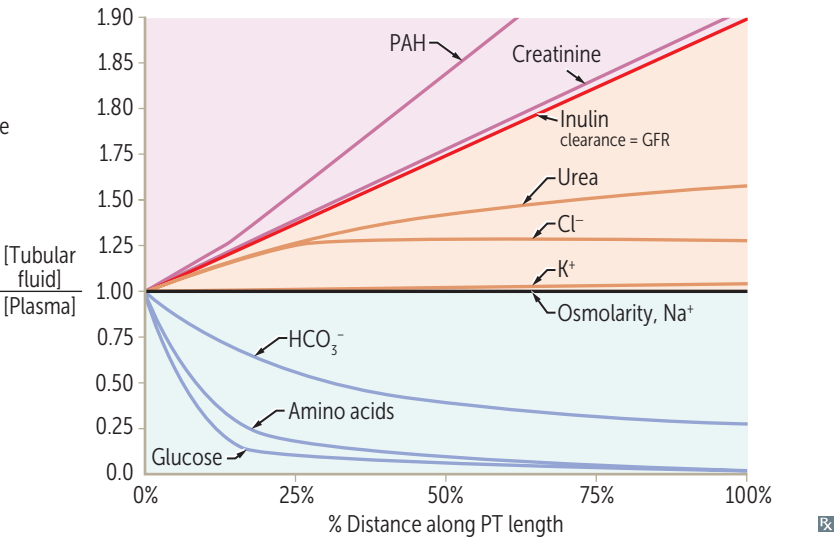
↑ ↓ = important differentiating feature.

Relative concentrations along proximal tubule

$[TF/P] > 1$
when solute is reabsorbed less quickly than water or when solute is secreted

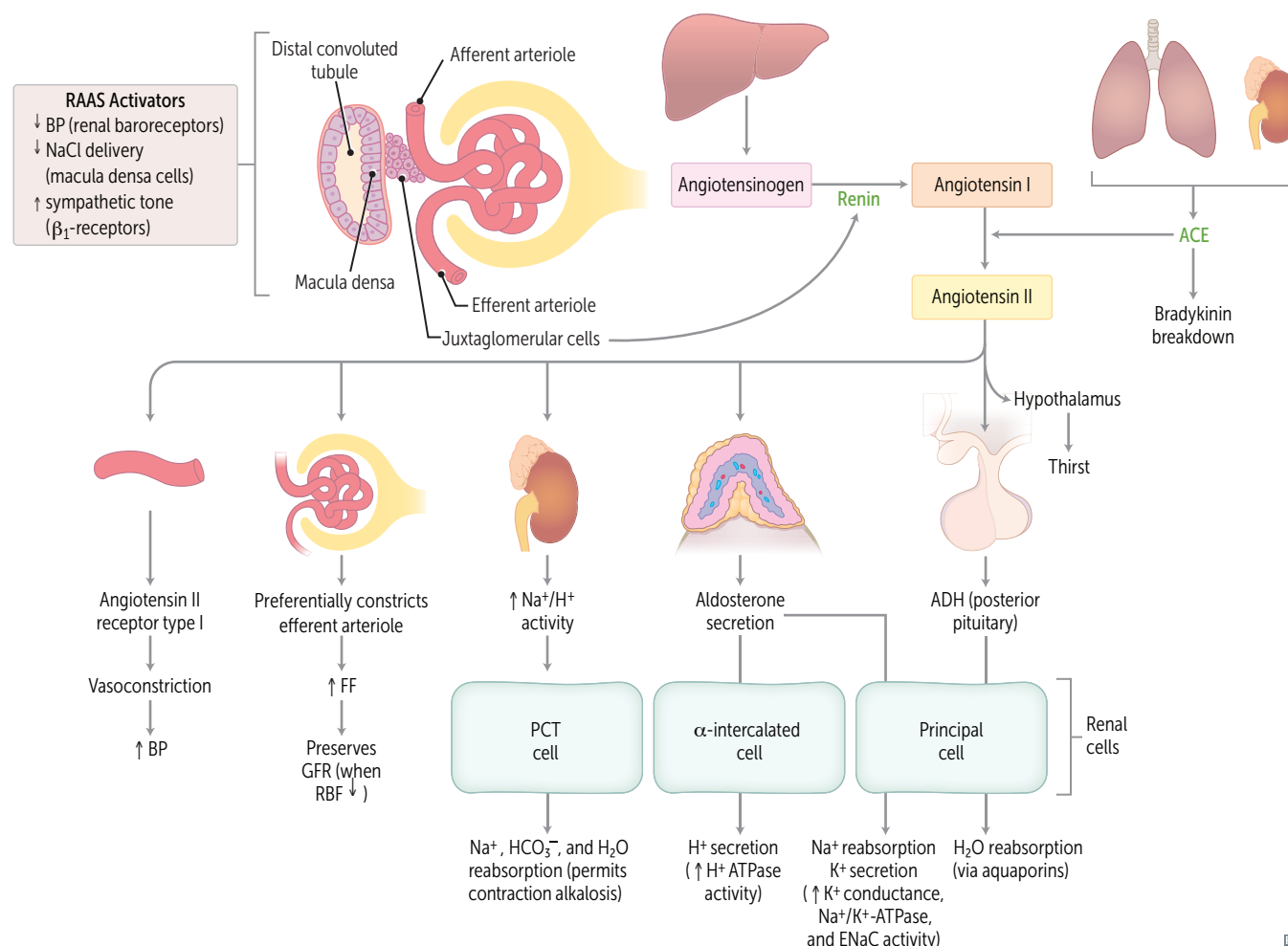
$[TF/P] = 1$
when solute and water are reabsorbed at the same rate

$[TF/P] < 1$
when solute is reabsorbed more quickly than water



Tubular inulin ↑ in concentration (but not amount) along the PT as a result of water reabsorption. Cl^- reabsorption occurs at a slower rate than Na^+ in early PCT and then matches the rate of Na^+ reabsorption more distally. Thus, its relative concentration ↑ before it plateaus.

Renin-angiotensin-aldosterone system



Renin	Secreted by JG cells in response to ↓ renal perfusion pressure (detected in afferent arteriole), ↑ renal sympathetic discharge (β_1 effect), and ↓ NaCl delivery to macula densa cells.
ACE	Catalyzes conversion of angiotensin I to angiotensin II. Located in many tissues but conversion occurs most extensively in the lung. Produced by vascular endothelial cells in the lung.
AT II	Helps maintain blood volume and blood pressure. Affects baroreceptor function; limits reflex bradycardia, which would normally accompany its pressor effects.
ANP, BNP	Released from atria (ANP) and ventricles (BNP) in response to ↑ volume; relaxes vascular smooth muscle via cGMP → ↑ GFR; ↓ renin → angiotensin-aldosterone inhibition. Dilates afferent arteriole, promotes natriuresis.
ADH (vasopressin)	Primarily regulates serum osmolality; also responds to low blood volume states. Stimulates reabsorption of water in collecting ducts. Also stimulates reabsorption of urea in medullary collecting ducts to maximize corticopapillary osmotic gradient.
Aldosterone	Primarily regulates ECF volume and Na^+ content; ↑ release in hypovolemic states. Responds to hyperkalemia by ↑ K^+ excretion.

Juxtaglomerular apparatus

Consists of mesangial cells, JG cells (modified smooth muscle of afferent arteriole), and the macula densa (NaCl sensor located at the DCT). JG cells secrete renin in response to ↓ renal blood pressure and ↑ sympathetic tone (β_1). Macula densa cells sense ↓ NaCl delivery to DCT → ↑ renin release → efferent arteriole vasoconstriction → ↑ GFR. Communication between JG cells and macula densa occurs via gap junctions.

JGA prevents short-term changes in GFR through autoregulation and maintains GFR long-term through regulation of the renin-angiotensin-aldosterone system. β -blockers ↓ BP by ↓ CO and inhibiting β_1 -receptors of the JGA → ↓ renin release.

Kidney endocrine/paracrine functions

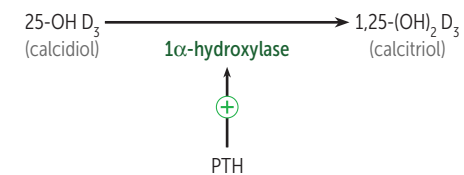
Erythropoietin

Released by interstitial cells in peritubular capillary bed in response to hypoxia.

Stimulates RBC proliferation in bone marrow. Administered for anemia secondary to chronic kidney disease. Adverse effect: ↑ risk of HTN in some individuals.

Calciferol (vitamin D)

PCT cells convert 25-OH vitamin D₃ to 1,25-(OH)₂ vitamin D₃ (calcitriol, active form). Increases calcium absorption in small bowel.



Prostaglandins

Paracrine secretion vasodilates afferent arterioles to ↑ RBF.

NSAIDs block renal-protective prostaglandin synthesis → constriction of afferent arteriole and ↓ GFR; this may result in acute kidney injury in low renal blood flow states.

Dopamine

Secreted by PT cells, promotes natriuresis. At low doses; dilates interlobular arteries, afferent arterioles, efferent arterioles → ↑ RBF, little or no change in GFR. At higher doses; acts as vasoconstrictor.

Hormones acting on kidney

Atrial natriuretic peptide

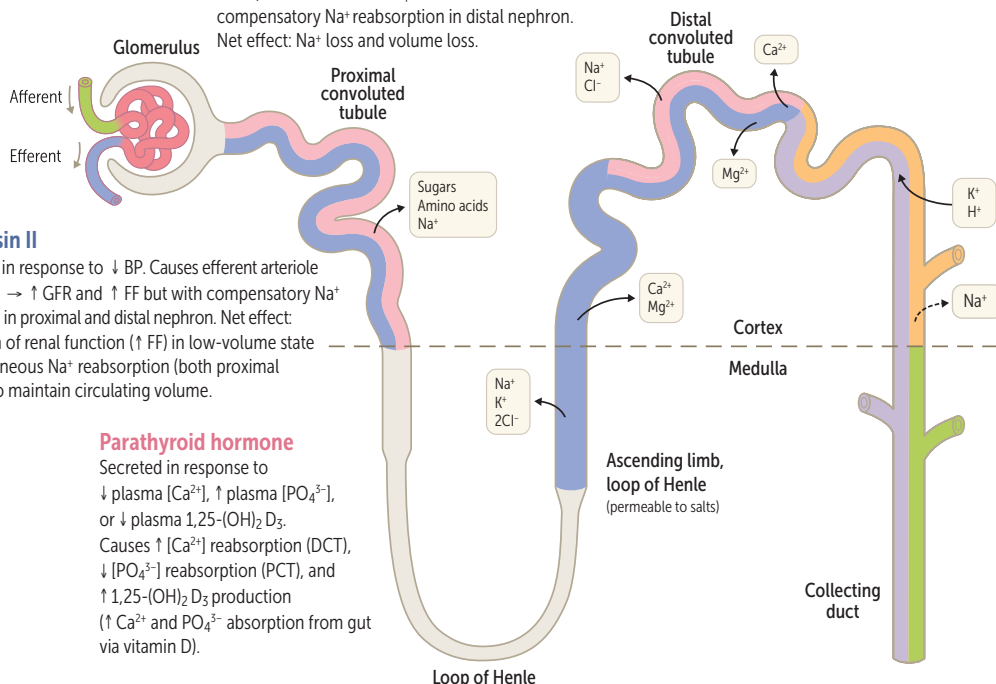
Secreted in response to ↑ atrial pressure. Causes indirect afferent arteriole dilation (through inhibition of NE). Causes ↑ GFR and ↑ Na^+ filtration with no compensatory Na^+ reabsorption in distal nephron. Net effect: Na^+ loss and volume loss.

Angiotensin II

Synthesized in response to ↓ BP. Causes efferent arteriole constriction → ↑ GFR and ↑ FF but with compensatory Na^+ reabsorption in proximal and distal nephron. Net effect: preservation of renal function (↑ FF) in low-volume state with simultaneous Na^+ reabsorption (both proximal and distal) to maintain circulating volume.

Parathyroid hormone

Secreted in response to ↓ plasma $[\text{Ca}^{2+}]$, ↑ plasma $[\text{PO}_4^{3-}]$, or ↓ plasma $1,25\text{-(OH)}_2\text{D}_3$. Causes ↑ $[\text{Ca}^{2+}]$ reabsorption (DCT), ↓ $[\text{PO}_4^{3-}]$ reabsorption (PCT), and ↑ $1,25\text{-(OH)}_2\text{D}_3$ production (↑ Ca^{2+} and PO_4^{3-} absorption from gut via vitamin D).



Aldosterone

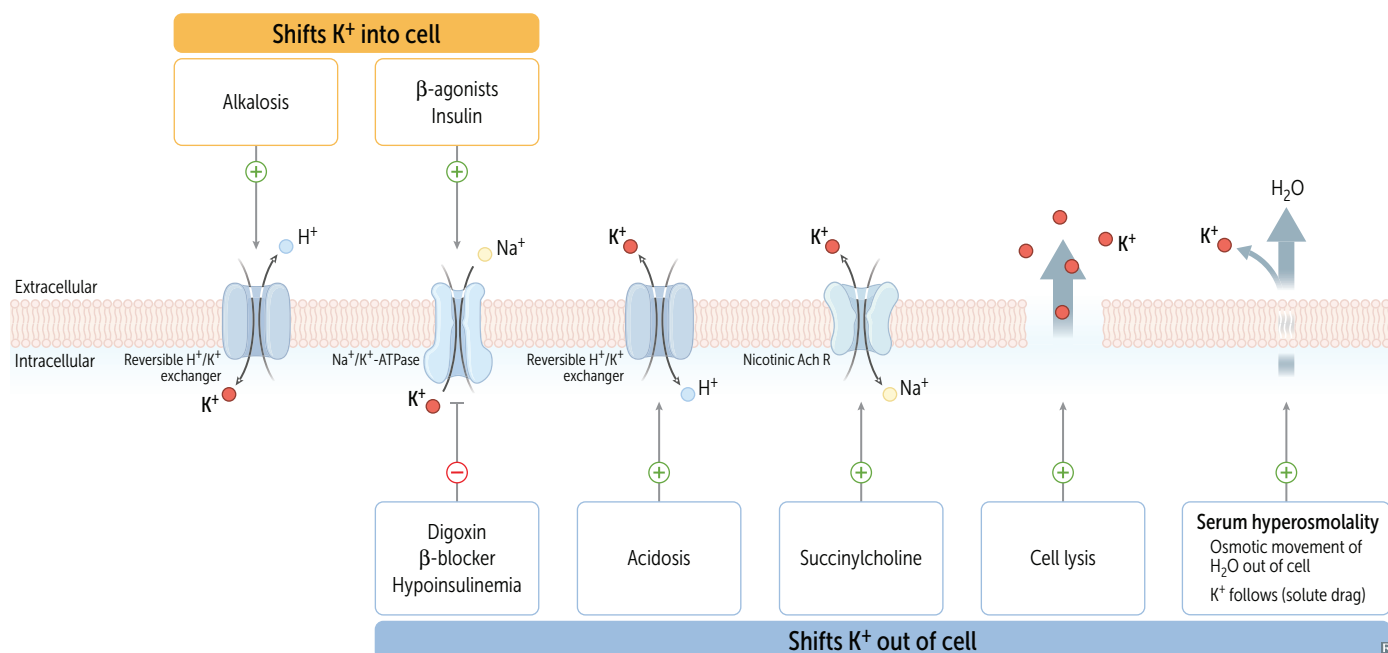
Secreted in response to ↓ blood volume (via AT II) and ↑ plasma $[\text{K}^+]$; causes ↑ Na^+ reabsorption, ↑ K^+ secretion, ↑ H^+ secretion.

ADH (vasopressin)

Secreted in response to ↑ plasma osmolarity and ↓ blood volume. Binds to receptors on principal cells, causing ↑ number of aquaporins and ↑ H_2O reabsorption. ↑ reabsorption of urea in medullary collecting ducts to maximize corticopapillary osmotic gradient.

Rx

Potassium shifts



Rx

Electrolyte disturbances

ELECTROLYTE	LOW SERUM CONCENTRATION	HIGH SERUM CONCENTRATION
Sodium	Nausea, malaise, stupor, coma, seizures	Irritability, stupor, coma
Potassium	U waves and flattened T waves on ECG, arrhythmias, muscle cramps, spasm, weakness	Wide QRS and peaked T waves on ECG, arrhythmias, muscle weakness
Calcium	Tetany, seizures, QT prolongation, twitching (eg, Chvostek sign), spasm (eg, Trousseau sign)	Stones (renal), bones (pain), groans (abdominal pain), throne s (↑ urinary output frequency), psychiatric overtones (anxiety, altered mental status)
Magnesium	Tetany, torsades de pointes, hypokalemia, hypocalcemia (when $[Mg^{2+}] < 1.0$ mEq/L)	↓ DTRs, lethargy, bradycardia, hypotension, cardiac arrest, hypocalcemia
Phosphate	Bone loss, osteomalacia (adults), rickets (children)	Renal stones, metastatic calcifications, hypocalcemia

Acid-base physiology Metabolic acid-base disorders cause HCO_3^- alterations. Respiratory acid-base disorders cause PCO_2 alterations.

	pH	PCO_2	$[HCO_3^-]$	COMPENSATORY RESPONSE
Metabolic acidosis	↓	↓	↓	Hyperventilation (immediate)
Metabolic alkalosis	↑	↑	↑	Hypoventilation (immediate)
Respiratory acidosis	↓	↑	↑	↑ renal $[HCO_3^-]$ reabsorption (delayed)
Respiratory alkalosis	↑	↓	↓	↓ renal $[HCO_3^-]$ reabsorption (delayed)

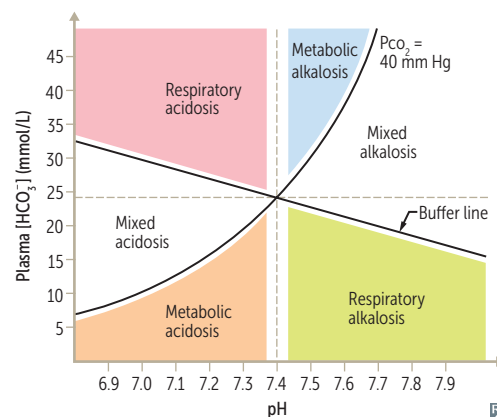
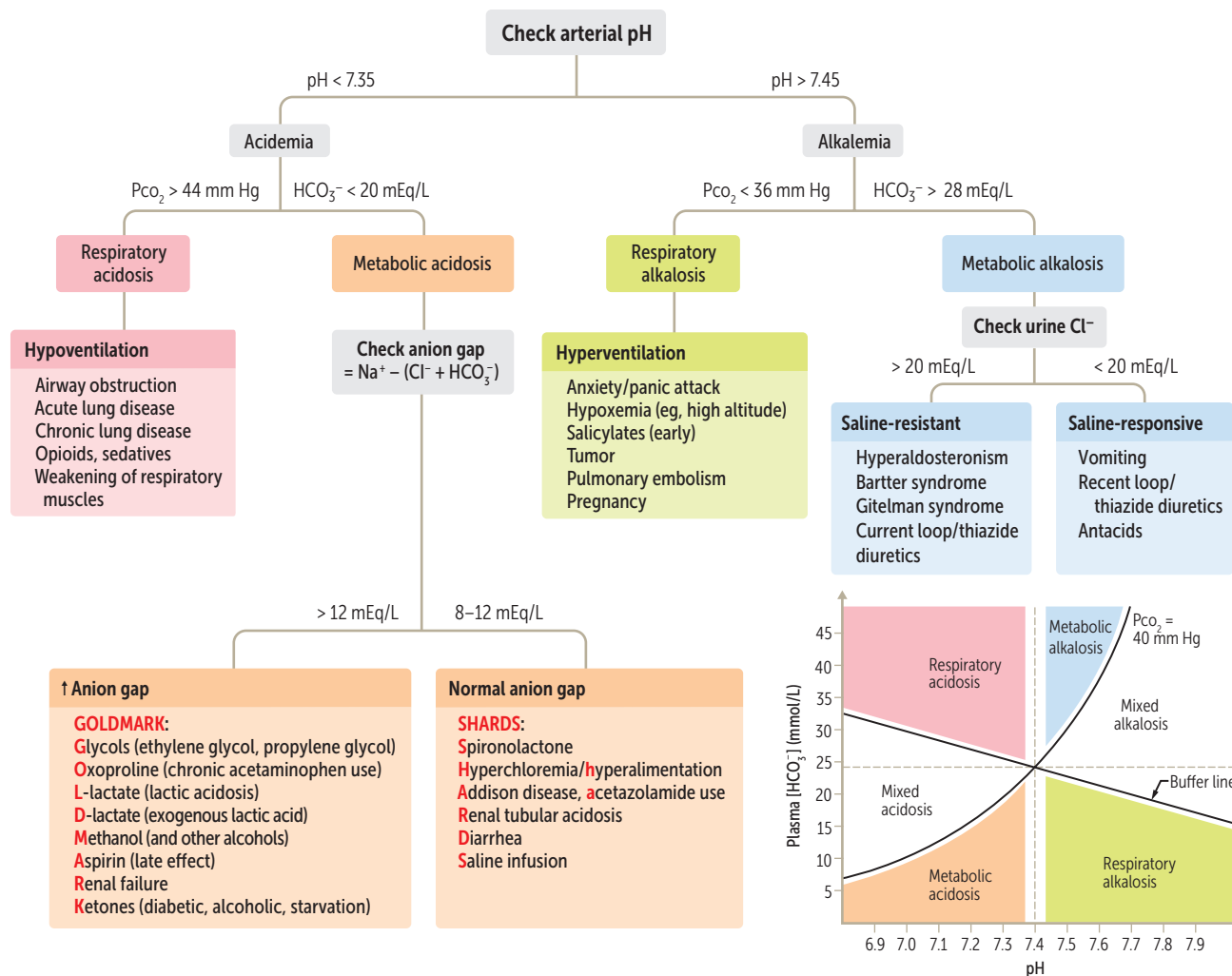
Key: ↓ ↑ = compensatory response.

$$\text{Henderson-Hasselbalch equation: } pH = 6.1 + \log \frac{[HCO_3^-]}{0.03 PCO_2}$$

Predicted respiratory compensation for a simple metabolic acidosis can be calculated using the Winters formula. If measured $PCO_2 >$ predicted $PCO_2 \rightarrow$ concomitant respiratory acidosis; if measured $PCO_2 <$ predicted $PCO_2 \rightarrow$ concomitant respiratory alkalosis:

$$PCO_2 = 1.5 [HCO_3^-] + 8 \pm 2$$

Acidosis and alkalosis



Renal tubular acidosis

	Distal renal tubular acidosis (RTA type 1)	Proximal renal tubular acidosis (RTA type 2)	Hyperkalemic tubular acidosis (RTA type 4)
DEFECT	Inability of α -intercalated cells to secrete $H(1)^+$ $\rightarrow$ no new HCO_3^- is generated $\rightarrow$ metabolic acidosis	Defect in PCT bi(2)carbonate (HCO_3^-) reabsorption $\rightarrow$ $\uparrow$ excretion of HCO_3^- in urine $\rightarrow$ metabolic acidosis Urine can be acidified by α -intercalated cells in collecting duct, but not enough to overcome $\uparrow$ HCO_3^- excretion	Hypoaldosteronism or aldosterone resistance; hyperkalemia $\rightarrow$ $\downarrow$ NH_3 synthesis in PCT $\rightarrow$ $\downarrow$ NH_4^+ excretion
URINE pH	> 5.5	< 5.5 when plasma HCO_3^- below reduced resorption threshold > 5.5 when filtered HCO_3^- exceeds resorptive threshold	Variable
SERUM K^+	$\downarrow$	$\downarrow$	$\uparrow$
CAUSES	Amphotericin B toxicity, analgesic nephropathy, congenital anomalies (obstruction) of urinary tract, autoimmune diseases (eg, SLE)	Fanconi syndrome, multiple myeloma, carbonic anhydrase inhibitors	$\downarrow$ aldosterone production (eg, diabetic hyporeninism, ACE inhibitors, ARB, NSAIDs, heparin, cyclosporine, adrenal insufficiency) or aldosterone resistance (eg, K^+ -sparing diuretics, nephropathy due to obstruction, TMP-SMX)
ASSOCIATIONS	$\uparrow$ risk for calcium phosphate kidney stones (due to $\uparrow$ urine pH and $\uparrow$ bone turnover related to buffering)	$\uparrow$ risk for hypophosphatemic rickets (in Fanconi syndrome)	

